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Reinforcement learning based UAV formation control in GPS-denied environment

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Abstract Highly accurate positioning is a crucial prerequisite of multi Unmanned Aerial Vehicle close-formation flight for target tracking, formation keeping, and collision avoidance. Although the position of a UAV can be obtained through the Global Positioning System (GPS), it is difficult for a UAV to obtain highly accurate positioning data in a GPS-denied environment (e.g., a GPS jamming area, suburb, urban canyon, or mountain area); this may cause it to miss a tracking target or collide with another UAV. In particular, UAV close-formation control in GPS-denied environments faces difficulties owing to the low-accuracy position, close distance between vehicles, and nonholonomic dynamics of a UAV. In this paper, on the one hand, we develop an innovative UAV formation localization method to address the formation localization issues in GPS-denied environments; on the other hand, we design a novel reinforcement learning based algorithm to achieve the high-efficiency and robust performance of the controller. First, a novel Lidar-based localization algorithm is developed to measure the localization of each aircraft in the formation flight. In our solution, each UAV is equipped with Lidar as the position measurement sensor instead of the GPS module. The k-means algorithm is implemented to calculate the center point position of UAV. A novel formation position vector matching method is proposed to match center points with UAVs in the formation and estimate their position information. Second, a reinforcement learning based UAV formation control algorithm is developed by selecting the optimal policy to control UAV swarm to start and keep flying in a close formation of a specific geometry. Third, the innovative collision risk evaluation module is proposed to address the collision-free issues in the formation group. Finally, a novel experience replay method is also provided in this paper to enhance the learning efficiency. Experimental results validate the accuracy, effectiveness, and robustness of the proposed scheme.

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1. Introduction

With the development of science and technology, there is an increasing demand on the intelligence and clustering of UAVs.^{1–3} The performance of a single UAV is restricted. For example, the survey area and inspection range are restricted when performing ground detection tasks.^{4–6} When performing transportation tasks, the capacity and transportation range will be limited.^{7–10} In addition, if a UAV fails in performing a task, the whole task is jeopardized. Multiple UAVs in close formation have attracted much attention in the research community^{11–13} because they can better explore search regions and dimensions. Close-formation flight involves flying in a formation geometry with a lateral spacing that does not exceed twice the wingspan. For a complex, dynamic flight environment, it is desirable for multiple UAVs in close formation to be capable of high-accuracy position measurement and disturbance rejection. High-accuracy localization and a stable controller are the most cited challenges for UAV close-formation tracking control in GPS-denied environments.^{2,14}

The problem of UAVs in close formation has attracted much attention in the research community. Various formation strategies relevant to UAV flocking have been developed, including the leader–follower strategy, distributed formation control, behavior-based method, and virtual formation structure. Each solution has its strengths and weaknesses. For example, the leader–follower strategy has a simple system structure which can be conveniently implemented, but it is overly dependent on the performance of the leader UAV, and has insufficient communication between the leader and followers. Yang et al.¹⁵ proposed an adaptive output feedback controller to address the leader-following consensus problem, where the action of a single UAV can reach consensus in formation flight. However, this method does not consider the influences of backlash and hysteresis on the control model. The scientist Ali et al.¹⁶ proposed an efficient adaptive control method addressing the transmission delay problem in the leader–follower formation. In another solution,¹⁷ the virtual leader technology is also utilized to provide a robust and effective formation controller. Yu et al.¹⁸ proposed a neural adaptive control algorithm to accelerate convergence and eliminate tracking errors in distributed formation structure. Wang et al.¹⁹ provided a switching methodology based controller to address the consensus problem in swarms. Liu et al.²⁰ optimized a non-linear mapping method and prescribed a controller to realize collision avoidance in the distributed formation structure. The primary challenge of the behavior-based algorithm is utilizing the behavior state of each aircraft in an optimal routine. The virtual formation structure achieves highly adaptive performance and has sufficient robustness because each vehicle can be transferred to a virtual leader or follower state. However, the virtual structure requires a high communication and computation capacity, which may not be easily implemented on a UAV in real-time applications.

We carefully designed a novel flight control system for UAV close formations in GPS-denied environments, whose framework is shown in Fig. 1. Each UAV is equipped with Lidar for coordinated localization. UAVs in formation are regarded as dynamic objects, and each UAV can measure the relative positions of surrounding UAVs by processing Lidar point cloud data. A Deep Deterministic Policy Gradient

(DDPG)-based control algorithm is proposed to achieve stability and robust performance in formation flight. The proposed strategy can measure the relative localization of the drones in a formation without using the GPS, import coordinated relative positions into the control module to guide vehicles in formation flying in a GPS-denied environment, and conduct close formation tracking missions with collision avoidance. To the best of our knowledge, we are the first to develop an Reinforcement Learning (RL)-based control methodology to address the Vertical Take-off and Landing (VTOL) aircraft close formation problem in GPS-denied environments. Our contributions are presented as follows.

- (1) This work provides the proof that UAV can execute the close formation flight in GPS-denied environments. Lidar-based solution is developed to measure relative positions in formation flying. Compared with traditional positioning solutions, the proposed Lidar-based navigation method can provide higher-quality relative localization and ensure better formation flight than a GPS sensor.
- (2) The novel relative localization method is developed to measure the position of each aircraft in formation flight. The k-means algorithm is implemented to estimate the center point positions of UAVs. A new formation position vector matching algorithm is proposed to match localization data with UAVs' number in formation, making the Lidar-based solution efficient and accurate in real-world scenario.
- (3) A novel reinforcement learning based formation control algorithm is proposed. We develop a Deep Reinforcement Learning (DRL)-based controller to obtain an optimal control policy through deep neural networks, which enables the UAV to be adapted in complicated formations. To avoid collisions, a new collision-risk evaluation algorithm is developed. Furthermore, the dynamic prioritized experience replay algorithm is proposed to enhance the learning efficiency and practical implementation.
- (4) The effectiveness and robustness of the designed system are verified on a high-fidelity hardware-in-the-loop simulation platform, and real-world flight experiments further show the performance of the proposed formation control system without using global positioning sensors. Our method can accurately measure relative positions in real situations and keep the formation with maneuvers of target UAVs.

The rest of this paper is organized as follows. Section 2 analyzes the problem state and its solutions. Section 3 introduces the relative localization strategy. Section 4 outlines the proposed control method. Experimental results are presented in Section 5. The conclusions are drawn in Section 6.

2. Problem state and its solutions

2.1. UAV localization method in GPS-denied environment

The performance of close formation relies much on GPS signals. The localization of a close formation can generally be measured in GPS-friendly situations, eliminating inherent

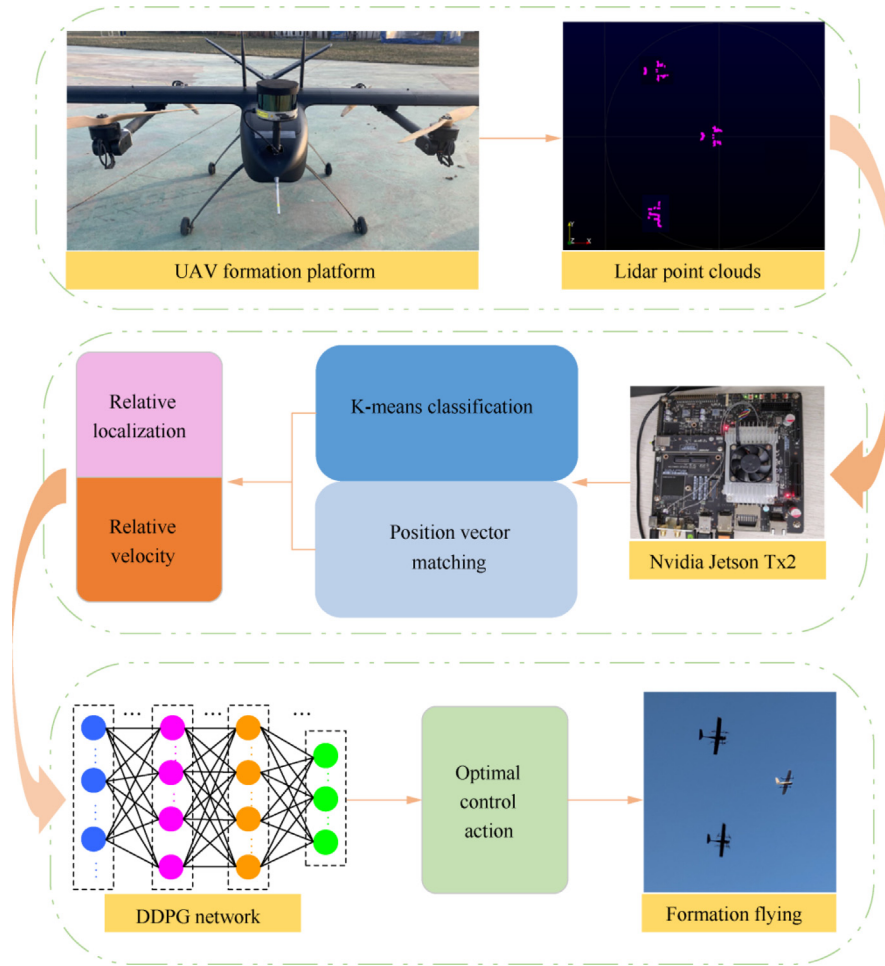


Fig. 1 Framework of proposed method.

instrumental errors in an Inertial Measurement Unit (IMU) such as of time drift, positioning, and speed.²¹ However, when GPS does not work normally, the single inertial navigation mode will accumulate errors in measurement due to its own inertial device attributes.^{22–25} The failure of the GPS receivers would lead to the failure of the close-formation mission. Overdependence on satellite navigation systems will lead to the decreased reliability and security of navigation and positioning. A UAV cannot accurately locate the long-distance flight process relying only on the inertial navigation system. Researchers have developed different methods to realize UAV flight in the absence of GPS. The visual-based relative localization algorithm²⁶ was designed for UAV navigation in GPS-denied environments, using terrain aided simultaneous localization and mapping algorithm to connect the UAV localization information and UAV's images. This method needs a standard aspect as an ancillary feature for measurement, and it is difficult to extend it to general uncontrolled situations. Guo et al.²⁷ designed a UAV localization method by fusing the information between the ultra-wideband measurement model and communication network in the UAV swarms. The drawback of this solution is that its accuracy is not stable in long-range and high-speed dynamic applications. Hemann et al.²⁸ designed an error-state Kalman filter with IMU bias estimation to maintain a high-certainty state estimate, and reduced the computation time to search over a global elevation

map. Lidar has been used to scan a sub-region of Digital Elevation Mode (DEM) to estimate the aircraft state. Shen et al.^{29–30} proposed an effective Lidar/visual-based method for place recognition and localization applications. Xu et al.³¹ developed a visual-inertial-UWB fusion solution which combines the camera and Ultra Wide Band (UWB) model for the relative localization problems. The Lidar/visual detection based relative localization methods are developed on the quadrotor sensing systems.^{32–33} A novel particle swarm based localization method is proposed to obtain the position of aircraft without using GPS sensors.³⁴ Experimental results demonstrated that performance in continual tracking decreased when there was a complicated background. Some common challenges of this method are that it requires high computation consumption and it has low positioning accuracy in real-world implementation. A single camera and Lidar fusion model were equipped on a UAV platform to achieve higher localization accuracy.²⁴

In this study, we firstly compared several popular localization methods, including Lidar-based method, ultrasonic-based method,³⁵ visual-based method,²⁶ and UWB-based method.²⁷ Table 1^{26,27,35} shows the comparison results in multiple aspects, including accuracy, detection range, inference time, and power. It can be seen that the proposed Lidar-based solution has the best performance in both accuracy and detection range, which are the most important factors for the close formation applications. And the other metrics of Lidar are also

Table 1 Localization methods' comparison results for formation application in GPS-denied environment.

Solution	Accuracy(m)	Range(m)	Inference time(ms)	Power(W)
Lidar-based	0.02	150	6	20
Ultrasonic-based ³⁵	0.10	15	0.3	1
Visual-based ²⁶	0.45	10	10	8
UWB-based ²⁷	0.21	40	3	4

appropriate to be implemented on the fixed wing UAV. Considering these advantages, in this work, we investigate and develop the Lidar-based localization solution for UAV formation control in GPS-denied environment. The detailed information is demonstrated in [Section 3](#).

2.2. Reinforcement learning based UAV formation control solutions

A high-precision, stable UAV control method is crucial to close-formation flight. Traditional control technology can ensure a certain accuracy of control. However, in close formation in GPS-denied environments, it faces many challenges due to the stability, robustness and precision requirements of the controller. A discrepancy exists between the simplified UAV model and the real dynamic plant, which may decrease the performance of close-formation flight. An RL-based method^{36–39} can realize the optimal control policy and output optimal control instructions by learning the real world, and RL technology has been employed to simulate a dynamic UAV model and improve control accuracy.^{40–43} Based on the deep reinforcement learning technology,⁴⁴ Pham et al.^{45–46} studied the autonomous navigation process of UAVs, and applied it to the autonomous target area coverage problem. The improved learning-based adaptive formation control method⁴⁷ decreases the probability of UAVs' collisions in complex environment. Hu et al.⁴⁸ proposed a novel cooperative control method based on the reinforcement learning theory, which makes the UAV swarms be adaptive to the actuators' fault in the formation flight. The relevant experience learning algorithm was developed to speed up the training phase of the controller.⁴⁹ An optimized multi-agent learning-based formation control method was developed.⁵⁰ The learning experiences of each agent can be used by sharing policy to improve the control accuracy. The most cited challenge of the learning-based solution is improving learning efficiency.

Inspired by the above methods, we investigate a learning-based system of UAV close-formation control that is practical and efficient for real-world applications. In this study, we develop the physical VTOL aircraft and show successful real-world formation experiments without using GPS signal. The detailed content can be seen in [Section 4](#) and [Section 5](#).

3. Relative localization strategy

For the close-formation mission of VTOL UAVs in GPS-denied environments, a key issue is to obtain the relative location of each aircraft. Solutions such as Lidar, visual measurement, ultrasonic sensors, and motion capture systems can measure the relative distances of devices without a GPS module. We developed a novel Lidar-based localization method for three reasons.

- (1) Reliability: vehicles in close formations move at high speeds; this requires stable, high-resolution measurement results for efficient guidance information. Lidar can obtain the highest localization accuracy and run stably while an aircraft moves at a high velocity with significant vibrations.
- (2) Measurement range: a UAV needs to sense other objects to avoid collisions. A single visual camera, ultrasonic sensor, and UWB module have a limited ranging area, which does not provide the optimal solution in VTOL close-formation tasks. Lidar has a large measurement range, and can provide surrounding localization for an aircraft in a high-speed formation.
- (3) A power-limited UAV onboard computer: this can well process Lidar data in real time.

Hence we propose a Lidar-based localization algorithm for UAV close formation, by which a UAV can measure the relative positions of its neighbors by processing point cloud data without GPS sensors. The collected data include the linear distance of a detected object, and the angles of the object relative to the Lidar on the horizontal plane and in the vertical direction. It should be noted that a follower UAV in this paper is equipped with Lidar to measure the position of its neighbor. We use the k-means algorithm to process and classify the original point cloud data to calculate the center point position of neighbor UAVs around followers. The point clouds handled by k-means can only know the position data where the UAVs exist, and cannot associate this with the number information of UAVs in formation. To solve this problem, we design a formation position vector matching algorithm to obtain the corresponding number of position data.

3.1. Formation point cloud process

The point cloud data obtained by UAV-equipped Lidar contain points of the UAV and those around it. The point cloud of the UAV should be filtered out. According to the installation position of Lidar on the vehicle and its fuselage size, we set a threshold D1 and filter out points with a distance less than that. We use a UAV with length 1.5 m and wingspan 3 m, with Lidar installed at a height of 0.15 m. According to the Lidar measurement principle and research experience, we set D1 = 2.5 m.

In the process of tracking and maintaining formation flight, the point cloud data from each UAV collected by Lidar are divided into several point clouds. Since vehicles need to maintain a certain distance in the formation, there is a distance between point cloud clusters, according to which the k-means algorithm is implemented to obtain the coordinate relative position information. K-means is an unsupervised clustering algorithm based on distance, which is suitable for

UAV formation positioning. All points of each cloud cluster are used to calculate its average and obtain the center points of clouds. Then the distance from each point to the center point is calculated. The classification of a point depends on the distance between it and the center point. We iterate in this way until the points of the same classification tend to be divided into the same category, and the difference among different categories is maximized. Finally, the location information of surrounding UAVs is obtained.

In formation flight, followers are equipped with Lidar to scan the relative positions of neighbors and obtain Lidar point clouds. The minimum distance between UAVs in the formation is $D2$. The total number of collected Lidar points is N . We randomly select one point p_1 and select another point p_2 in the remaining point cloud. Similarly, we select a point p_3 , and we may ensure the distance between p_3 and p_1, p_2 is bigger than $D2$, and so on, until all points meeting this condition are selected. The UAV formation coordinate point set $P = \{p_1, p_2, \dots, p_k\}$ is generated, where k is the number of UAVs detected by Lidar.

The k center points are selected as the initial center points, and the Euclidean distance between each point and each center point is calculated. A point is assigned to the nearest center point to generate k groups of point clouds, and we obtain k classification point clouds. Then new center points are generated by calculating the mean of the k clouds mentioned above. We iterate the above steps; when the Euclidean distance between the new center and the previous center is less than $D3$, the iteration is stopped and the position of the latest center point is used to represent the center coordinates of the UAV. In our experiments, $D3$ was set to 0.25 m.

3.2. Formation position vector matching algorithm

According to the above k-means-based point cloud process method, we can only get the center coordinates of other UAVs around followers, but we cannot get the corresponding UAV formation number of these center coordinates. To address this problem, we designed a new Formation Position Vector Matching (FPVM) algorithm to associate the center position with the number of UAVs in a formation. The relative position of each UAV in formation can be known through the preset formation map. Due to errors during flight, the actual position of a drone may deviate from its predetermined position. It is supposed that the actual position of a drone is within a circle with the preset position as the center and position deviation R as the radius. From this, the pre-formation relation position vector of a UAV can be calculated. Its direction is from the position of follower to another UAV. Lidar can measure the relative localization between UAVs. We define the relative localization vector measured by Lidar as the Lidar position vector. Its direction is from the position of the follower to the position of another UAV detected by Lidar. In formation flight, a vehicle can obtain $n-1$ pre-formation position vectors and k Lidar position vectors, where n is the number of vehicles in the formation, and k is the number of vehicles detected by Lidar-equipped followers.

By comparing the lengths and angles of the vectors, the best matching pair of vectors can be selected, and the UAV number corresponding to the position vector of Lidar can be known. Therefore, the complete formation relative localization

information can be obtained. An instance of FPVM is demonstrated in Fig. 2, from which it can be seen that, for a specific UAV in formation, $(xp_{ij}, yp_{ij}, zp_{ij})$ is a pre-formation position and $(xl_{ij}, yl_{ij}, zl_{ij})$ is a Lidar position. We match the two vectors according to their lengths, angles, and directions. If the length and angle of vectors are matching at the same time, we consider that pre-formation position vector is matching the Lidar position vector, and assign the UAV number of pre-formation position vector to Lidar position vector. The method is formulated as follows.

- (1) Calculate the difference ε between the lengths of two vectors,

$$\varepsilon = \sqrt{xp_{ij}^2 + yp_{ij}^2 + zp_{ij}^2} - \sqrt{xl_{ij}^2 + yl_{ij}^2 + zl_{ij}^2} \quad (1)$$

where K_1 is the threshold criterion of length matching. If $\varepsilon < K_1$, two vectors are considered to have the same length. We set K_1 by experience, and here we set $K_1 = 0.8$.

- (2) Calculate the angle differences between two vectors. Calculate the projection angle of pre-formation position vectors and Lidar position vectors on the XOY plane:

$$\gamma_1 = \left| \arctan\left(\frac{yl_{ij}}{xl_{ij}}\right) - \arctan\left(\frac{yp_{ij}}{xp_{ij}}\right) \right| \quad (2)$$

the projection angle of two vectors on the XOZ plane,

$$\gamma_2 = \left| \arctan\left(\frac{zl_{ij}}{xl_{ij}}\right) - \arctan\left(\frac{zp_{ij}}{xp_{ij}}\right) \right| \quad (3)$$

and the projection angle of two vectors on the YOZ plane,

$$\gamma_3 = \left| \arctan\left(\frac{zl_{ij}}{yl_{ij}}\right) - \arctan\left(\frac{zp_{ij}}{yp_{ij}}\right) \right| \quad (4)$$

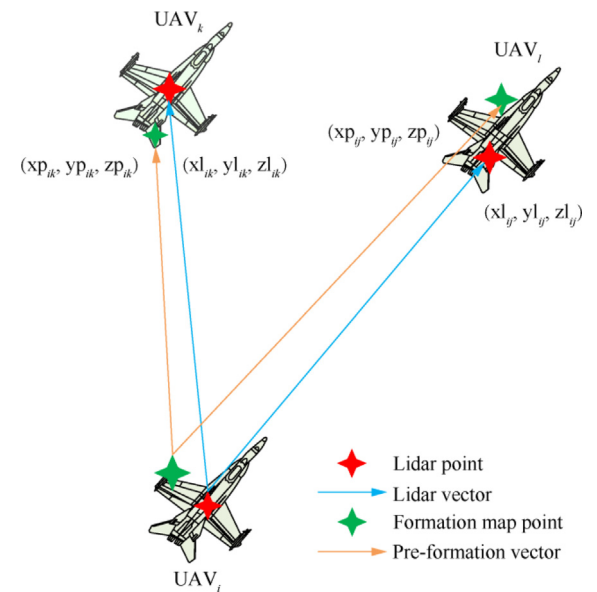


Fig. 2 Formation position vector matching method.

We use σ to denote the sum of γ_1 , γ_2 and γ_3 . If $\sigma < K_2$, two vectors are considered to have the same angle. We set $K_2 = 0.3$. If conditions Eqs. (1) and (2) match, we consider that the pre-formation position vector matches the Lidar position vector, and we assign the UAV number of the pre-formation position vector to the Lidar position vector. Then we can clearly know the relative localization and number information of other UAVs around the followers. The relative velocity between a follower and a detected UAV is estimated according to the following:

$$\begin{cases} \Delta v_{xt} = \frac{x_{ij}^t - x_{ij}^{t-1}}{\Delta T} \\ \Delta v_{yt} = \frac{y_{ij}^t - y_{ij}^{t-1}}{\Delta T} \\ \Delta v_{zt} = \frac{z_{ij}^t - z_{ij}^{t-1}}{\Delta T} \end{cases} \quad (5)$$

where ΔT is the time interval of Lidar sampling. The complete relative localization and relative velocity information is fed into the formation controller to maintain a stable UAV formation in a GPS-denied environment.

4. DRL-based formation control algorithm

A deep reinforcement learning based algorithm is applied to the UAV formation control task. The deep deterministic policy gradient method is a deep reinforcement learning solution combining value and policy iteration. Fig. 3 shows the architecture of the proposed reinforcement learning control method, which enables agent to achieve the optimum policy for an infinite state space and action space, so that UAVs perform better in the learning process for formation tracking tasks.

4.1. Actor-critic network

The deep deterministic policy gradient is a policy-based deep reinforcement learning algorithm that works with continuous state and action spaces. Policy-based reinforcement learning methods aim to directly search the optimal policy π^* , which provides a feasible framework for continuous control. DDPG

utilizes the actor-critic paradigm, adopts an experience playback mechanism and a separate target network, reduces the correlation between data, and increases stability and robustness. To approximate a deterministic policy (Actor) and Q function (Critic), the actor-critic structure, which is composed of an actor network and critic network, is applied in the DDPG algorithm.

The actor network module observes the state of the environment, uses the neural network to obtain the selection probability of action, and completes the interaction process between agent and environment. The rewards calculated from the interaction are utilized to maintain and update the action selection policy of agent. The critic network evaluates the value of the environmental state and action. To be specific, the critic network estimates the value of the actor network module by observing the state and action, and its output can guide the action selection policy of the actor network. It can be seen that there are two networks for the actor-critic algorithm, and the actor and critic networks need to construct their own neural networks. The artificial neural network of the actor network module realizes the refecton from the observation state to the action selection policy of the agent, and its training process must consider the error of the critic network module. The artificial neural network of the critic network obtains scores through the observation of the environmental state and action, and maps them to the corresponding score. The soft update method is applied to update the parameters of target networks, and can be expressed as

$$\begin{cases} \theta^{Q'} \leftarrow \tau \theta^Q + (1 - \tau) \theta^{Q'} \\ \theta^{\mu'} \leftarrow \tau \theta^{\mu} + (1 - \tau) \theta^{\mu'} \end{cases} \quad (6)$$

where τ is a soft update factor, which is generally small, ranging from 0.1 to 0.001; Q means the state-action value of the current critic network, and Q' is the state-action value of the current critic network; μ denotes the action value of the current actor network, and μ' denotes the action value of the target actor network.

To increase the degree of randomness in the learning process and enhance the learning coverage, DDPG adds random

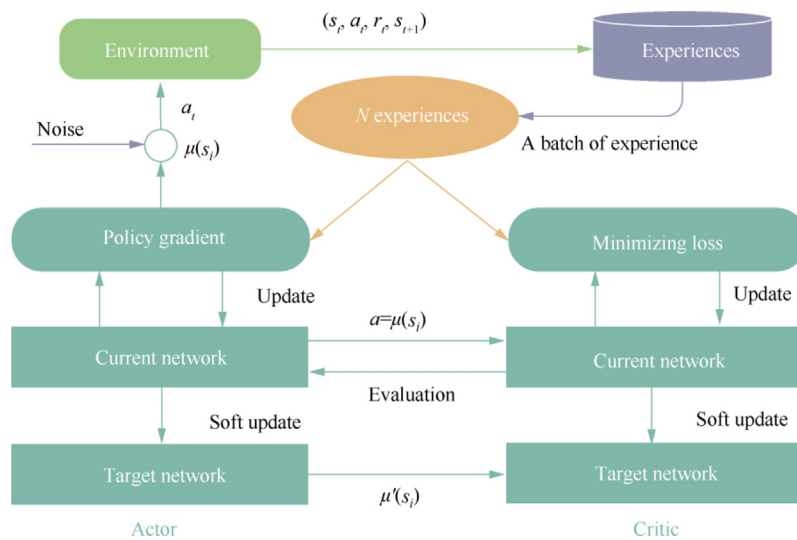


Fig. 3 Architecture of deep deterministic policy gradient control module.

noise N_t to selected action a . The optimal action generated from the policy network is defined as

$$a_t = \pi(s_t | \theta_t^\mu) + N_t \quad (7)$$

The critic is a network that evaluates the performance of all actors, and is optimized by minimizing the loss function,

$$L = \frac{1}{N} \sum_i (y_i - Q(s_i, a_i | \theta^Q))^2 \quad (8)$$

where $y_i = r(s_i, a_i) + \gamma Q(s_{i+1}, a_i | \theta^Q)$. To find the optimal solution, the loss function of the actor network can be defined as

$$\nabla_J(\theta^\mu) = \frac{1}{N} \sum \nabla_a Q(s_i, a_i | \theta^Q) \nabla_{\theta^\mu} \mu(s_i | \theta^\mu) \quad (9)$$

4.2. Collision risk evaluation map

We present a novel collision risk evaluation map for UAV formations to avoid collisions with obstacles and other moving UAVs during flight. The UAV domain^{2,9,10} represents the flight area where collisions should be avoided. The UAV domain is a sphere whose center is at the center point of the aircraft. Its radius is R_c , which we define as 3 m. We denote a random position around a UAV in the formation region as p , and the collision awareness factor is

$$C_p = \begin{cases} 1 - \frac{D}{R_c} & D \leq R_c \\ 0 & \text{otherwise} \end{cases} \quad (10)$$

where D is the Euclidean distance between arbitrary position p and the center of the aircraft. Then the global collision risk awareness factor can be derived by calculating the collision risk of each UAV in the formation group. In this work, the collision event of each UAV is considered an independent event. The global collision risk awareness is

$$C_g = 1 - \prod_j (1 - C_p^j) \quad (11)$$

where C_p^j is a collision risk evaluated by the j th UAV in the formation at position p . We construct a collision risk awareness map based on the global collision risk assessment results. The collision risk map M is an $L \times L \times H$ -dimension vector

centered at the leader UAV, as shown in Fig. 4, where the grayscale value of a pixel indicates the global collision risk awareness results. The collision risk map contains the collision risk information during the formation flight, which will be used as the input of the DRL model to enhance the effectiveness of collision avoidance.

4.3. UAV formation control model

We use a multi-target model instead of a leader-follower model to start and keep a UAV formation. Hence each UAV has a specific target. For example, in a “V” formation, UAV2 and UAV3 track UAV1 to maintain the formation. The target of UAV4 is UAV2. More target-follower relationships in different formation geometry are shown in Table 2. We divide the maneuvering process of a UAV formation tracking flight mission into two stages. In the guidance stage, a follower UAV is guided to approach the target UAV. In the formation maintenance stage, when a follower catches a target, it maintains a certain formation, i.e., it maintains a fixed distance from the target, and the speed and direction of UAVs should be the same in the second stage. The DDPG formation tracking control model is demonstrated as follows.

In practice, the formulation of the state and action spaces, as well as the design of the reward function, determine the speed of convergence and even the possibility of divergence of the reinforcement learning algorithm. A reasonable model design can help to accelerate the learning process and avoid divergence. In the proposed model, the state is

$$S = \{d_x, d_y, d_z, v_x, v_y, v_z, \phi, \theta, \psi, M\} \quad (12)$$

where $\{d_x, d_y, d_z\}$ consists of the relative distance in three directions from followers to the target at time t ; v_x, v_y , and v_z are relative velocities between the follower and target with respect to the Map Point (MP) in three directions at time t ; $\{\phi, \theta, \psi\}$ consists of the attitude angles of vehicles, i.e., the roll, pitch, and yaw, respectively; M is a collision-avoidance assessment map that presents the evaluation of the collision risk between the aircraft and its neighbors in the formation. A VOTL UAV with four rotors is used in our study, with action space

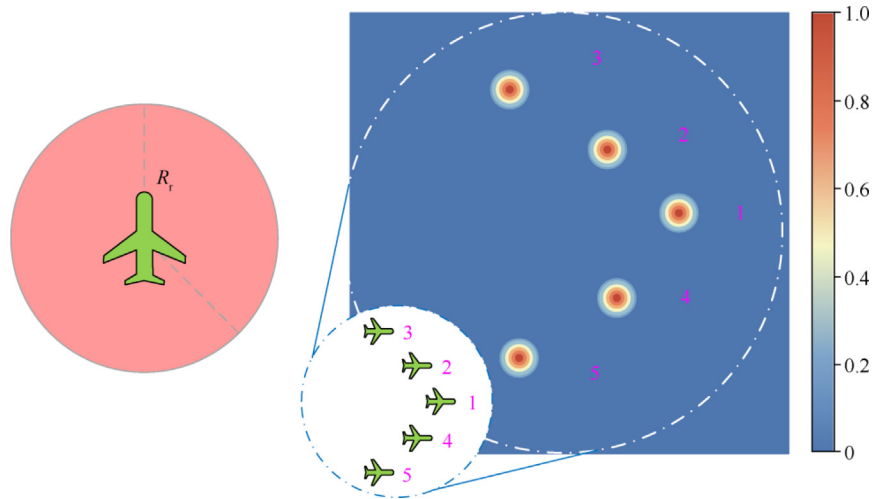


Fig. 4 Left: Example of UAV domain. Right: Visualization of collision risk evaluation map in formation.

Table 2 Target-follower relationship mapping.

“V” formation		“O” formation		Linear formation	
Follower	Target	Follower	Target	Follower	Target
UAV2	UAV1	UAV2	UAV1	UAV2	UAV1
UAV3	UAV1	UAV3	UAV2	UAV3	UAV2
UAV4	UAV2	UAV4	UAV2	UAV4	UAV3
UAV5	UAV3	UAV5	UAV3	UAV5	UAV4
UAV6	UAV4	UAV6	UAV4	UAV6	UAV5
UAV7	UAV5	UAV7	UAV5	UAV7	UAV6
UAV8	UAV6	UAV8	UAV6	UAV8	UAV7

$$A = \{\delta_{t_1}, \delta_{t_2}, \delta_{a_1}, \delta_{a_2}, \delta_{r_1}, \delta_{r_2}, U_1, U_2, U_3, U_4\} \quad (13)$$

where δ_t is the tilt angle, δ_a is the aileron deflection, δ_r is the rudder deflection, and $\{U_1, U_2, U_3, U_4\}$ represents four rotors in the vehicle.

A properly designed reward function is crucial in a DDPG model. An effective reward function can realize faster convergence and better performance. We analyze the characteristic of the whole stage of formation flying. To reach the optimal performance, we design a reward function,

$$\begin{cases} r_{\text{dis}} = -\lambda_1 (|d_x - \delta_x| + |d_y - \delta_y| + |d_z - \delta_z|) \\ r_{\text{con}} = \begin{cases} 0 & \xi > \xi_0 \\ -\lambda_2 |\xi_{t+1} - \xi_t| & \xi \leq \xi_0 \end{cases} \\ r_{\text{vel}} = \begin{cases} 0 & \xi > \xi_0 \\ -\lambda_3 |V_t - V_t| & \xi \leq \xi_0 \end{cases} \end{cases} \quad (14)$$

where δ_x, δ_y and δ_z denote the desired relative distances in three directions; ξ_0 is a certain Euclidean distance between the follower and target UAV in tracking maintenance; ξ is the real-time Euclidean distance between followers and targets; ξ_t is the relative distance at time t , $\xi_t = \sqrt{d_x^2 + d_y^2 + d_z^2}$; V_t is the relative velocity at time t , calculated as $V_t = \sqrt{v_x^2 + v_y^2 + v_z^2}$.

The purpose of r_{dis} is to guide the followers to approach the target UAV as soon as possible, r_{con} can act as a constraint to ensure there is no big shift in relative distance in the formation maintenance stage, and r_{vel} acts to make the UAVs keep the same velocity after a follower captures the target. Hence the total reward function is

$$r_{\text{total}} = r_{\text{dis}} + r_{\text{con}} + r_{\text{vel}} \quad (15)$$

This component is implemented to allow a UAV to dynamically adjust its flight status with high efficiency and precision in formation flight.

4.4. Dynamic prioritized experience replay algorithm

In the conventional DDPG algorithm, the initial action has high randomness, and the network parameters update with insufficient guidance, which makes convergence difficult. It takes a long time to train an appropriate control strategy. Based on previous work,^{38,39} the experience replay sample has various significance for the RL training process. Samples with high importance contain more information, which can accelerate convergence and make the training process more

efficient in the beginning periods. Samples with low importance are also helpful in the learning phase because they can increase the diversity and enhance the robustness of the model. We developed an innovative dynamic prioritized experience replay method that automatically adjusts the probability of a sample according to its importance and sampling times. The importance item

$$Wr_i = |TD_i| R_i + \alpha \quad (16)$$

evaluates the significance of the sample, where TD_i is the temporal-difference error of the i th learning sample, and

$$R_i = \lg(|r_i| + 1) + 1 \quad (17)$$

is the normalized representation of the current reward r_i . To decrease the probability of the experience when it has been sampled too many times, the sampling frequency item is considered in the importance item, which is given as

$$Fs_i = e^{-(\text{num}_i)^\alpha} \quad (18)$$

where num_i is the sampling times of the experience, and α is a hyper-parameter. Hence the synthetic importance item is defined as

$$Ws_i = Wr_i + \kappa Fs_i \quad (19)$$

where κ is a hyper-parameter used to balance the significant item and sampling times. Then the probability of sampling the experience is

$$P_i = \frac{Ws_i^\beta}{\sum_i Ws_i^\beta} \quad (20)$$

Algorithm 1 demonstrates how the DPER method works in the experience replay process.

Algorithm 1. Dynamic prioritized experience replay algorithm

1. **Input:**
2. Batch size of replay memory N .
3. **Output:**
4. N training samples.
5. Initialize the replay memory according to Eq. (19)
6. **for** i to N **do**
7. Sample a transition from the replay memory according to Eq. (19)
8. Calculate TD error of transition
9. Update sample weight according to Eq. (19)
10. **end for**

5. Formation flight experiments

We discuss the experimental configuration and results. The proposed reinforcement learning control algorithm was implemented in Python using TensorFlow.⁴⁴ We trained the proposed formation control model in virtual formation flight environments based on a self-developed flight simulator, consisting of an environment module, multi-type UAV model, physical engine, sensor modules, rendering interface, and interface API. To make the model better adapt to the real environment, we executed the training process in a real-world formation flight to fine-tune the network parameters. We utilized the parameters obtained in a virtual environment, and implemented the model on an Nvidia Jetson Tx2 onboard flight control computer. In each episode of real-world training, UAVs conducted close-formation flight tasks to finish training episodes. When a vehicle deviated from the desired formation trajectory, the safety drone operator took over the flight-control system and ended the episode. The state data recorded by the flight log were used to update parameters and optimize the control policy. After an amount of fine-tuning training in real environments, the proposed model could achieve the desired effectiveness and robustness.

5.1. Numerical experiments and analysis

The flight space was 10 km long, 8 km wide, and 150 m high. The maximum velocity of UAVs was 30 m/s. In each episode, a UAV formation started in a random position and executed formation control. In the training process configuration, we used a Dynamic Prioritized Experience Replay (DPER) mechanism to accelerate convergence and improve learning efficiency. Using the same experimental setup, we compared three DRL-based formation control algorithms with regard to the learning efficiency: Deep Reinforcement Learning Dynamic Prioritized Experience Replay (DRL-DPER), Deep Reinforcement Learning Prioritized Experience Replay (DRL-PER),³⁸ and DRL.³⁷ Fig. 5 illustrates the average reward per episode, which increased more rapidly for the proposed method because its dynamic adjustment mechanism could be more efficient when selecting training samples with high importance. The curves of the average reward per episode of DRL-PER and DRL almost overlap, which means that they have roughly the same performance. In the initial training stage, the return curve of DRL-PER grows slightly faster than that of DRL, which indicates that competitive networks can learn control strategies more effectively by optimizing the sampling mechanism. Compared with the other two methods, the proposed algorithm could more effectively sample high-priority samples in the training phase, which allowed it to rise rapidly and then stabilize with a good reward performance.

The designed formation control algorithm was implemented on the Nvidia Jetson, which was equipped on the aircraft as a flight controller. We carried out comprehensive formation flight experiments to verify the performance of the proposed method. The experimental platform was a VTOL UAV, with four electric propellers and a wingspan of 3 m. Table 3 presents its basic parameters. A VLP-16 Lidar was mounted on the head of the UAV, as shown in Fig. 6(a). Fig. 6(c) shows the visualization results of Lidar measurement data in the formation scenario. The Lidar's detection range

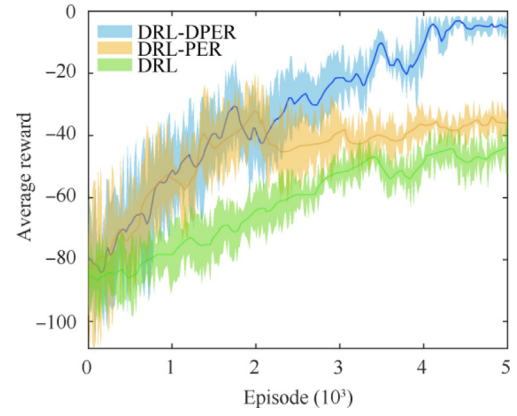


Fig. 5 Average reward via episode.

Table 3 Parameters of UAV platform.

Parameter	Value
Payload (kg)	5.2
Wingspan (m)	3
Length (m)	1.5
Cruise time (h)	2.5
Maximum cruise speed (m/s)	30

was 150 m, and its range accuracy was ± 2 cm. The proposed algorithm, sensing, and communication were executed on the Nvidia Jetson, which had been installed inside the belly of the VTOL.

5.2. Hardware in loop test

To further analyze the effectiveness and feasibility of the designed formation control system, Hardware-in-the-Loop (HITL) experiments were conducted based on our self-developed flight simulator. As shown in Fig. 6(b), the HITL test system consists of a flight control computer, simulation platform, and ground control station. The designed reinforcement learning based formation control algorithm was implemented on the flight control computer, which processed control action to navigate the vehicles flying in the appropriate formation. The simulator ran on a simulation computer, which provided flight information during the formation. The flight control computer interacted with simulation environments and ground stations through a serial server. The ground control station was applied to monitor the real-time flying states of UAVs.

As shown in Fig. 7, the performance of formations of 5 and 25 vehicles based on the proposed method was experimentally verified. The preset trajectory was generated in a ground station, and a reference path file was fed into the controller of each drone. Virtual Lidar sensors measured the relative positions in formation flights instead of using a GPS signal. Followers used the control strategy learned in the numerical simulation environment to complete the task of following the leader. In the formation flight test of five vehicles, the forma-

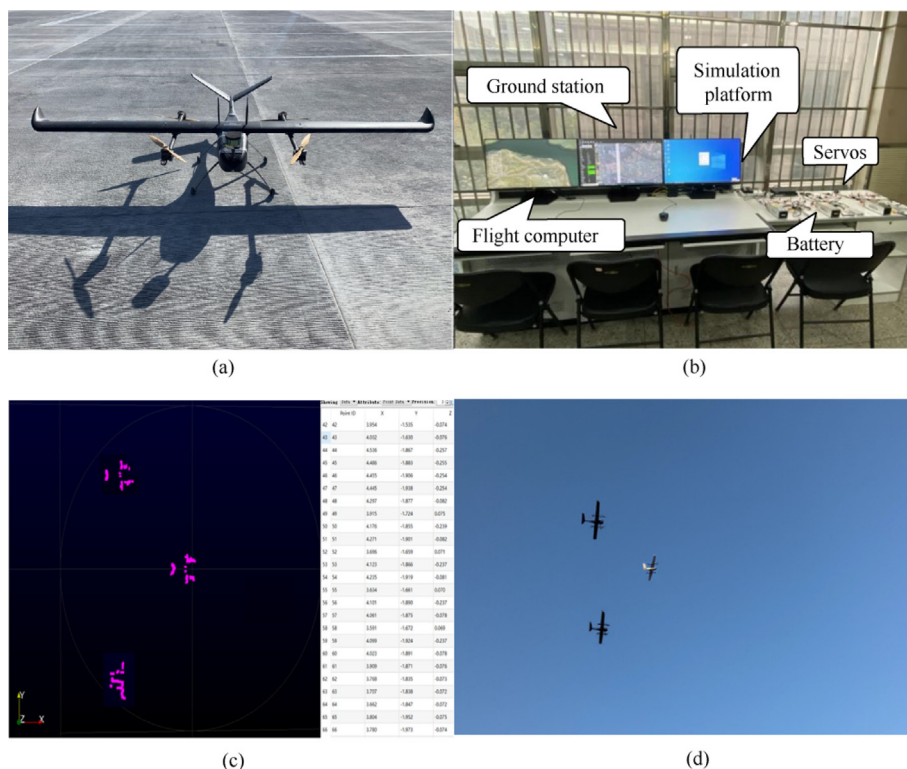


Fig. 6 UAV formation experimental platform.

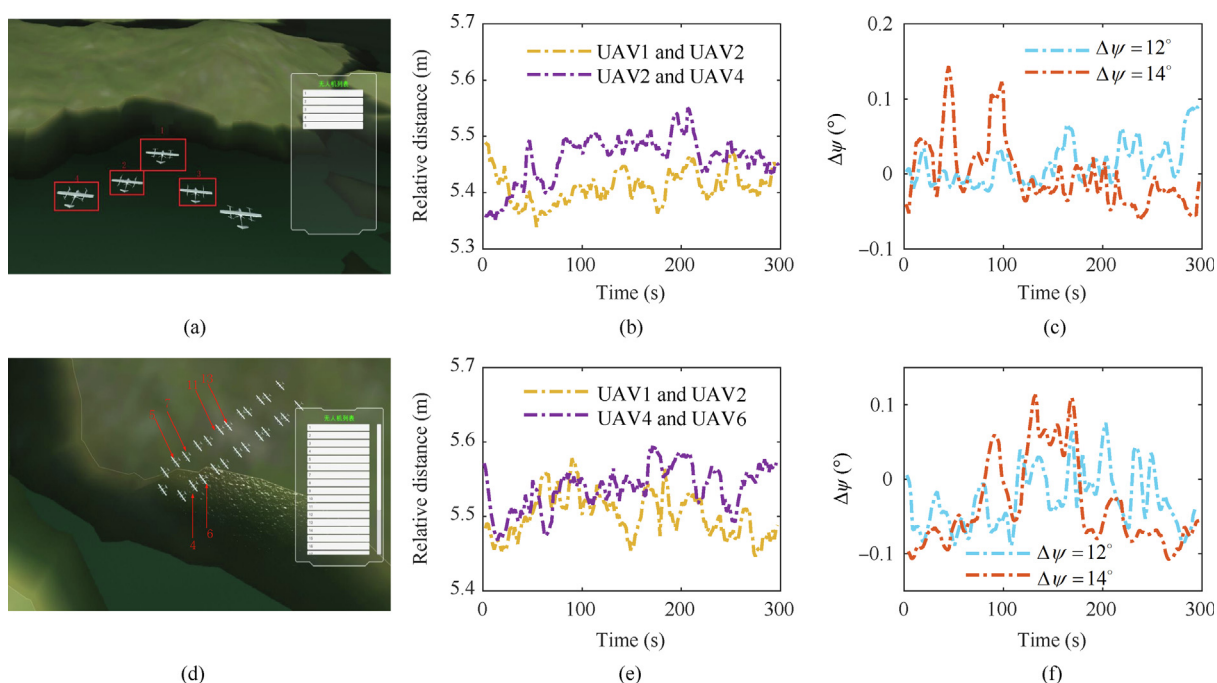


Fig. 7 HITL simulation. First row: Formation flight with 5 UAVs. Second row: Formation flight with 25 UAVs.

tion flew in a V geometry, and the followers were able to accurately track the leader in formation flight with the designed method. Fig. 7(b) and Fig. 7(c) illustrate the relative distance and relative yaw angle between UAV1 and UAV2, as well

as the relative distance between UAV2 and UAV4. In the test of 25 vehicles, flying in another geometry, Fig. 7(e) and Fig. 7(f) show the relative movement relationships between UAV1 and UAV2, and between UAV4 and UAV6, respec-

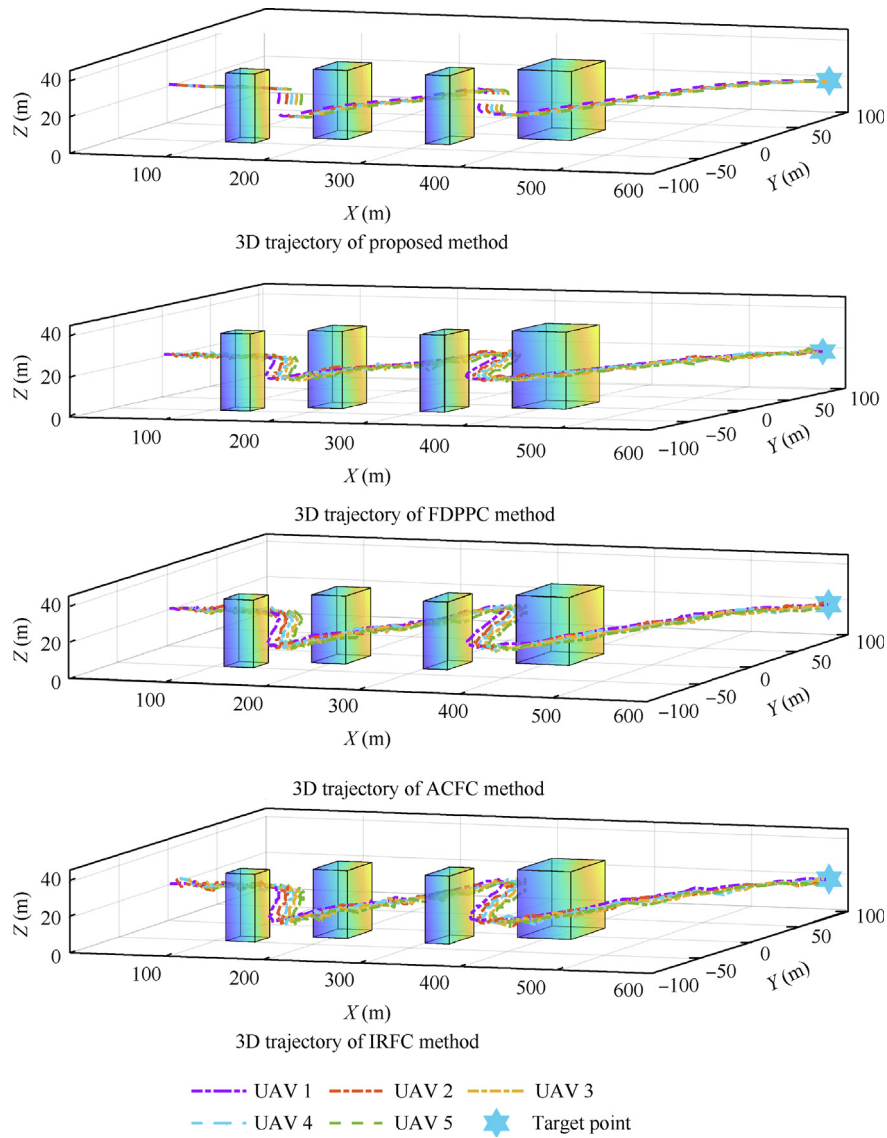


Fig. 8 3D trajectory results of formation flight in obstacle environment.

Table 4 Control errors of four methods in formation flight simulation.

Method	POS RMSE(m)	VEL RMSE(m)	ACC RMSE(m)	ATT RMSE(m)
Proposed method	0.2593	0.2708	0.2337	0.8231
FDPPC ¹	0.6234	0.6067	0.6712	1.6569
ACFC ⁶	1.4686	1.3649	1.2970	2.8526
IRFC ¹¹	1.8616	1.5176	1.4158	3.0165

tively. In HITL experiments, the proposed system worked well in multi-UAV formation flight, and each vehicle was able to keep the appropriate position with respect to the other UAVs.

5.3. Comparison experiments

To further verify the efficiency and robustness of the proposed method, we conducted the comparison experiments in regards to the formation control performance with several state-of-the-

art techniques, including the Fully Distributed Prescribed Performance Control (FDPPC) method,¹ Adaptive Consensus-based Formation Control (ACFC) method⁶ and Integrated Robust Formation Control (IRFC) method.¹¹ Fig. 8 shows the 3D simulated results of the different formation control algorithms. It can be seen from Fig. 8 that the proposed method is more efficient to make the UAV group fly towards the target position and avoid the collision with external environmental obstacles, and has the less fluctuations during the formation flight compared with other State of the

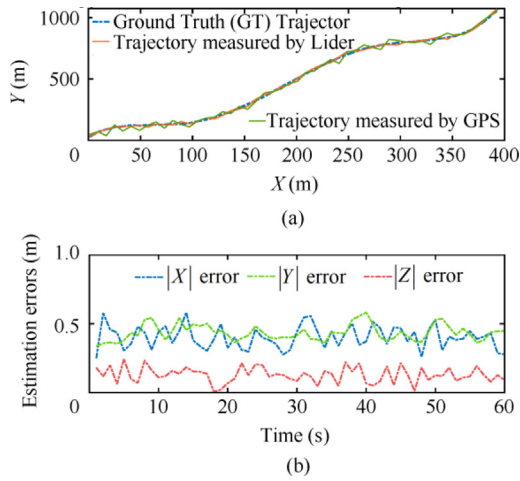


Fig. 9 Flight state recorded on UAV2.

Art (SOTA) methods. The Root Mean Square Error (RMSE) indicator is used to quantitatively assess the tracking control performance of the proposed method. M is defined as the samples size, and e_i denotes the position tracking error. The corresponding formulations are given as: $\text{RMSE} = \sqrt{\frac{1}{M} \sum_{i=1}^M e_i^2}$.

The numerical error analysis is presented in Table 4^{1,6,11}. As shown in Table 4, the RMSE metric of the proposed method in multiple channels are the smallest among other

three latest controllers. Moreover, the position errors of the proposed method are also significantly smaller than other methods. The reason is that the proposed method is able to achieve faster and more accurate response to external obstacles. In the training phase, the designed RL-based solution has learned appropriate control policy and the capability of collision avoidance. The comparison results verify that the proposed method achieves better robustness and faster response performance compared with other latest methods.

5.4. Formation flight experimental results

We verified the proposed strategy through real-world formation experiments, gradually increasing the number of vehicles in formation. Our technology routine is demonstrated as follows.

We verified the proposed strategy on three VTOL vehicles, where UAV1 was a target, and UAV2 and UAV3 were followers tracking it in flight. The formation is shown in Fig. 6(d), where the three UAVs achieve the expected equilateral triangle formation with side length 6 m, and maintain the formation with the movements of the target UAV. Fig. 6(c) shows the visualization of Lidar point clouds in formation flight. The control of these UAVs was set in a global coordinate frame with the x-axis aligned with magnetic East, and the Y-axis with magnetic North. Vehicles took off in multi-rotor mode at the home position and transitioned to fixed-wing mode, and formation flight began. The maximum wind speed was 6 m/s.

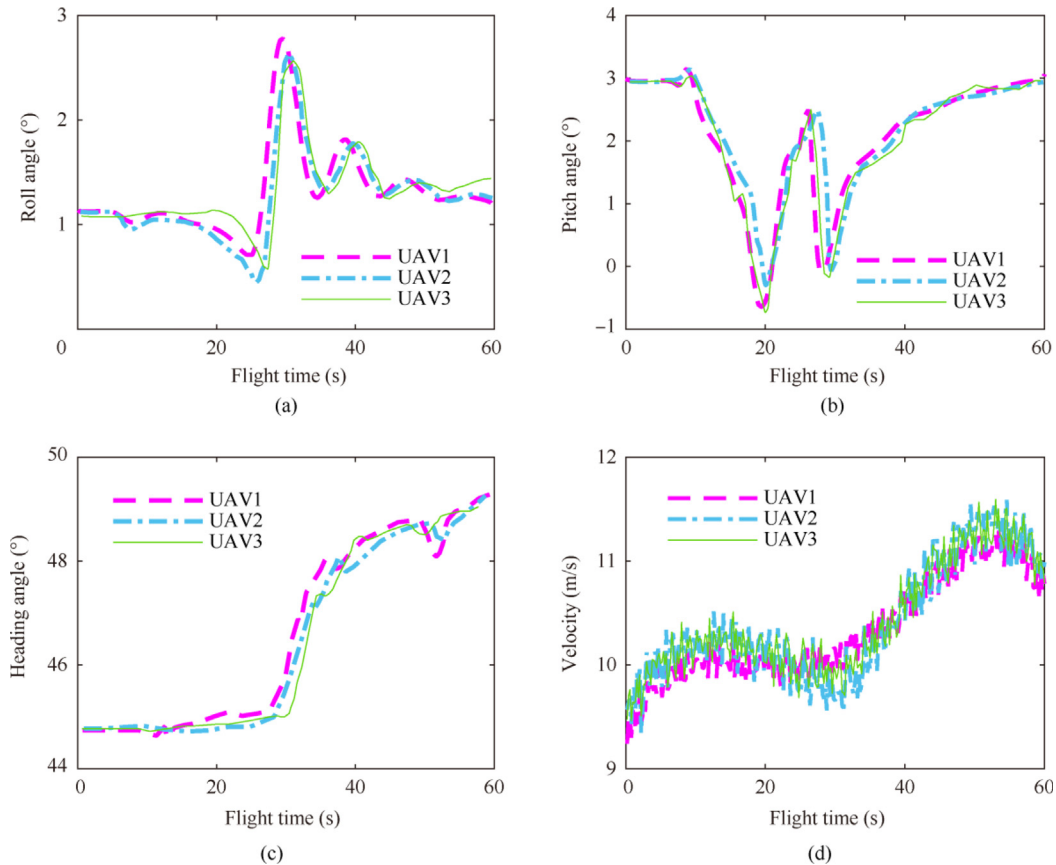


Fig. 10 Attitude angle and velocity of 3 vehicles.

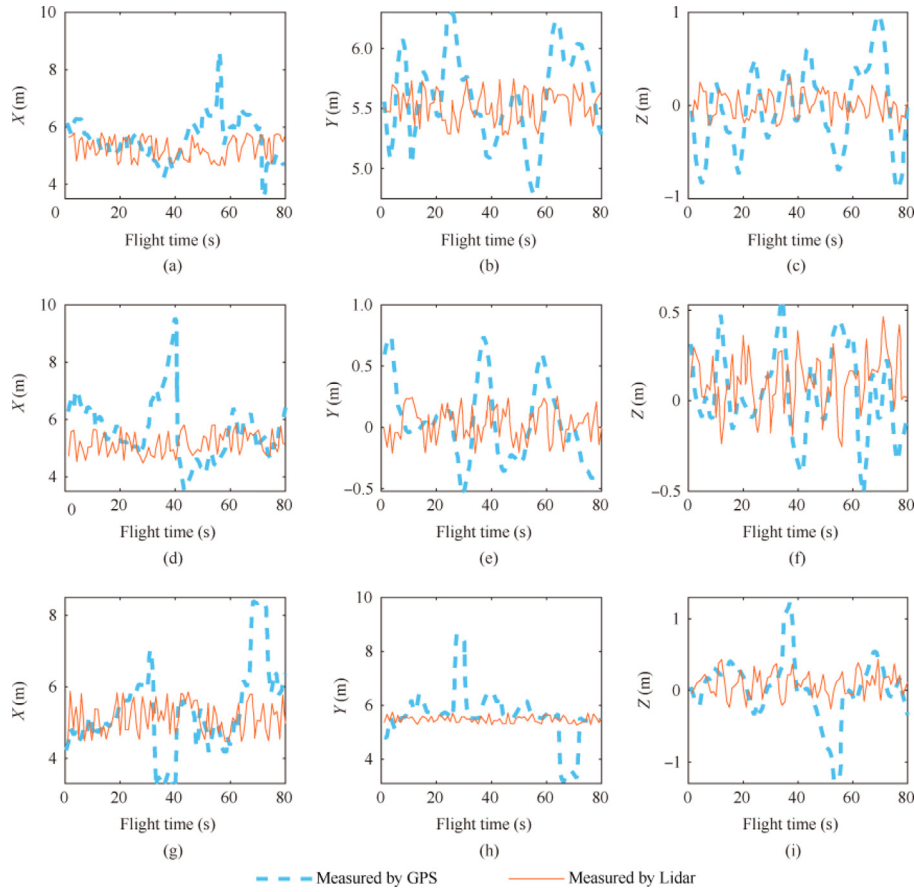


Fig. 11 Relative localization in formation flight.

Vehicles executed the formation control strategy to start flying in the triangle geometry. When keeping formation, each follower's velocity gradually tended to the final cruising velocity of the target UAV, and the heading angle tended toward its final value, which helped to keep a stable formation and precise geometry. To quantitatively verify the designed relative localization algorithm, we imply the Real Time Kinematic (RTK) module on the aircraft for the ground truth, and record the GPS data for comparison. It should be mentioned that all data from RTK and GPS were only used for comparative analysis in real-world outdoor flight, and not for UAV control. Then we compared the position performance among different position sensors. We used neither RTK nor GPS data for control; the relative position for the controller was directly calculated from Lidar. The flight state of each vehicle was recorded to demonstrate the proposed formation control algorithm.

The plot of the overhead 2D part of the flight trajectory recorded on UAV2 is presented in Fig. 9(a). The proposed localization method is more accurate than the GPS method, and comparable to RTK results. Fig. 9(b) shows the estimation errors compared with ground-truth values, where it can be seen that absolute estimation errors of relative localization in the X and Y directions are both less than 0.6 m, and the estimation error in the Z direction is less than 0.25 m. To analyze the origins of errors, we tested the estimation errors in a static environment on the ground. The estimation errors in three axes were all less than 0.05 m. This is because there is vibration in a dynamic formation flight environment, which may lead to

Lidar detection errors. Furthermore, the UAV body's surface area in the vertical plane is less than that in its horizontal plane. Thus, it is clear that the estimation errors in the vertical direction are less than those in the horizontal direction. However, the proposed Lidar-based localization strategy has achieved its goal, and the proposed strategy can work for formation flight missions in GPS-denied environments.

In the formation mission, it is important to control the flight state of three vehicles to maintain consistency. As can be seen from Fig. 10, the attitude angles of UAV2 and UAV3 can accurately track the attitude angles of UAV1 in formation flight. Fig. 10(d) shows velocity curves when followers are tracking the target UAV, and those of followers always coincide with UAV1 while its flight speed is being maneuvered. The experimental results demonstrate that the proposed relative localization strategy and DDPG control algorithm achieve promising performance for a three-UAV formation in a GPS-denied environment.

As a further test of the robustness and effectiveness of the proposed method, we executed an eight-UAV formation flight task. Videos of our real-world formation experiments are available at <https://youtu.be/HgtUlzqFDfY>. The experiments were similar to those with the three-UAV formation. To measure the performance of formation control, we equipped a GPS and barometer on aircraft, whose data were only used for a comparison with the proposed localization method. Lidar was used to detect the relative positions of UAVs in formation, and a DDPG controller was used to make UAVs reach the

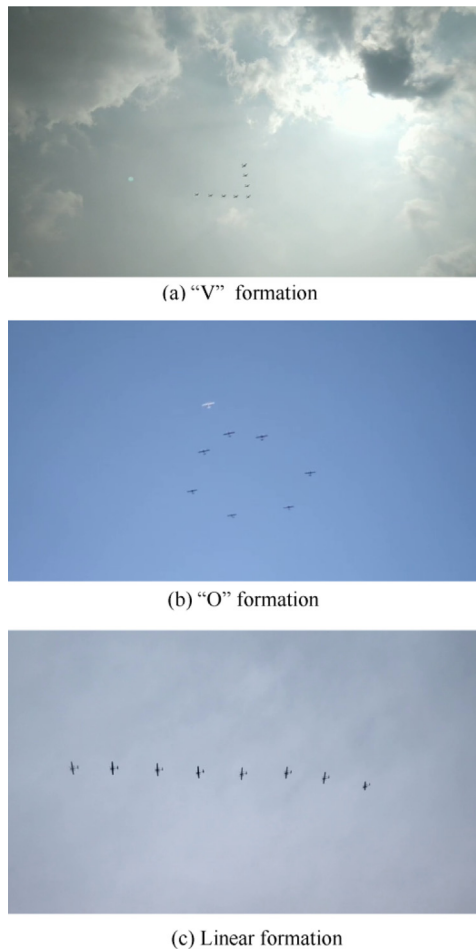


Fig. 12 Eight UAV formation experiments.

desired formation. We designed "V" "O" and linear formations. To illustrate the effectiveness of the proposed formation control method, we used relative localization data recorded in formation flight as evaluation criteria. We compared the relative localization estimated by Lidar and other traditional positioning methods. In traditional localization methods, position data are measured by GPS. Figs. 11(a)-(c) shows the relative localization recorded on UAV2 in the "V" formation. It can be seen that the vehicle was able to maintain a stable formation with a relative distance below 6 m along the X and Y axis. The mean relative distance along the Z axis remained at zero. The "O" formation relative localization recorded on UAV3 is shown in Figs. 11(d)-(f), and the linear formation relative distance is shown in Figs. 11(g)-(i). It can be seen that the Lidar-based localization method is more stable than the other positioning sensors. To better illustrate the formation control performance of the proposed method, the maximum error indicator and RMSE indicator are used in the formation flight experimental results analysis. Based on the real-world flight data in the formation experiments, the maximum error of the proposed method in the position channel is 0.35 m which is significantly smaller than the sphere radius value. And the RMSE indicator of the proposed method in the position channel is 0.22 m. It can be seen from the maximum error indicator

and the RMSE indicator of the experiments results that the proposed formation control method can work efficiently in the fixed-wing close formation scenario and keep the collision avoidance in the formation flight. Fig. 12 presents the scenarios of the eight-UAV formation. The experimental results show that the proposed strategy performs well for close-formation flight, and the localization data measured by Lidar are more accurate than those by traditional positioning sensors.

6. Conclusions

In this study, we proposed a cooperative relative localization methodology for close-formation flight in GPS-denied environments, where each vehicle in formation could measure the relative localization of its neighbors by processing Lidar point cloud data instead of using a GPS sensor. The novel relative localization algorithm is developed to measure the position of UAV in the formation. To estimate the center points of UAVs, a k-means-based point cloud method was developed. An innovative position vector matching algorithm is proposed to obtain the complete relative localization information in formation. The estimated inter-UAV distances and relative velocity were fed into the designed reinforcement learning based formation control module to generate optimal actions to control vehicles to start and keep flying in formation in a specific geometry. To avoid collisions in formation flight, a novel collision risk evaluation map was developed. A dynamic prioritized experience replay algorithm was proposed to improve training efficiency. A series of experiments, including simulation tests, hardware-in-the-loop experiments and comparison experiments with the latest methods were conducted to verify the theoretical elements, and real-world flight experiments further showed positive results. Our method overcame the difficulty of precisely localizing UAVs in formation flight without global positioning data, and provided robust and effective performance of formation control in complex flight situations. The method offered a highly efficient and stable solution for close UAVs in formation without GPS in practical engineering tasks. The combination of the proposed system with a visual navigation system will be investigated in our future work.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

1. An BH, Wang B, Fan HJ, et al. Fully distributed prescribed performance formation control for UAVs with unknown maneuver of leader. *Aerosp Sci Technol* 2022;**130**:107886.
2. Yan C, Wang C, Xiang XJ, et al. Deep reinforcement learning of collision-free flocking policies for multiple fixed-wing UAVs using local situation maps. *IEEE Trans Ind Inform* 2022;**18**(2):1260–70.
3. He A, Gao HG, Zhang SS, et al. Full mode flight dynamics modelling and control of stopped-rotor UAV. *Chin J Aeronaut* 2022;**35**(10):95–105.
4. Chen LL, Liu ZB, Dang QQ, et al. Robust trajectory tracking control for a quadrotor using recursive sliding mode control and nonlinear extended state observer. *Aerosp Sci Technol* 2022;**128**:107749.
5. Ma BD, Liu ZB, Jiang FH, et al. Vehicle detection in aerial images using rotation-invariant cascaded forest. *IEEE Access* 2019;**7**:59613–23.
6. Muslimov TZ, Munasypov RA. Consensus-based cooperative control of parallel fixed-wing UAV formations via adaptive backstepping. *Aerosp Sci Technol* 2021;**109**:106416.
7. Li H, Wang J, Han CW, et al. Leader-follower formation control of multiple UAVs based on ADRC: Experiment research. *2021 4th IEEE international conference on industrial cyber-physical systems (ICPS)*. Piscataway: IEEE Press; 2021. p. 558–65.
8. Huang Y, Meng ZY. Bearing-based distributed formation control of multiple vertical take-off and landing UAVs. *IEEE Trans Contr Netw Syst* 2021;**8**(3):1281–92.
9. Liu YC, Bucknall R. Path planning algorithm for unmanned surface vehicle formations in a practical maritime environment. *Ocean Eng* 2015;**97**:126–44.
10. Singh Y, Sharma S, Sutton R, et al. A constrained A* approach towards optimal path planning for an unmanned surface vehicle in a maritime environment containing dynamic obstacles and ocean currents. *Ocean Eng* 2018;**169**:187–201.
11. Zhang JL, Zhang P, Yan JG. Distributed adaptive finite-time compensation control for UAV swarm with uncertain disturbances. *IEEE Trans Circuits Syst I* 2021;**68**(2):829–41.
12. Zhang QR, Liu HHT. UDE-based robust command filtered backstepping control for close formation flight. *IEEE Trans Ind Electron* 2018;**65**(11):8818–27.
13. Peng ZH, Wang D, Li TS, et al. Output-feedback cooperative formation maneuvering of autonomous surface vehicles with connectivity preservation and collision avoidance. *IEEE Trans Cybern* 2020;**50**(6):2527–35.
14. Liu C, Jiang B, Zhang K. Adaptive fault-tolerant H-infinity output feedback control for lead-wing close formation flight. *IEEE Trans Syst Man Cybern* 2020;**50**(8):2804–14.
15. Yang ZC, Zheng SQ, Liu F, et al. Adaptive output feedback control for fractional-order multi-agent systems. *ISA Trans* 2020;**96**:195–209.
16. Ali ZA, Israr A, Alkhamash EH, et al. A leader-follower formation control of multi-UAVs via an adaptive hybrid controller. *Complexity* 2021;**2021**:1–16.
17. Ali ZA, Han ZG. Multi-unmanned aerial vehicle swarm formation control using hybrid strategy. *Trans Inst Meas Contr* 2021;**43**(12):2689–701.
18. Yu YJ, Guo J, Ahn CK, et al. Neural adaptive distributed formation control of nonlinear multi-UAVs with unmodeled dynamics. *IEEE Trans Neural Netw Learn Syst* 2022;**99**:1–7.
19. Wang CP, Wang JQ, Wu P, et al. Consensus problem and formation control for heterogeneous multi-agent systems with switching topologies. *Electronics* 2022;**11**(16):2598.
20. Liu BJ, Li AJ, Guo Y, et al. Adaptive distributed finite-time formation control for multi-UAVs under input saturation without collisions. *Aerosp Sci Technol* 2022;**120**:107252.
21. Chen LL, Liu ZB, Gao HG, et al. Robust adaptive recursive sliding mode attitude control for a quadrotor with unknown disturbances. *ISA Trans* 2022;**122**:114–25.
22. Liu WL, Li ZX, Sun SS, et al. Design a novel target to improve positioning accuracy of autonomous vehicular navigation system in GPS denied environments. *IEEE Trans Ind Inform* 2021;**17**(11):7575–88.
23. Tang YZ, Hu YC, Cui JQ, et al. Vision-aided multi-UAV autonomous flocking in GPS-denied environment. *IEEE Trans Ind Electron* 2019;**66**(1):616–26.
24. Lezki H, Yetik İŞ. Localization using single camera and lidar in GPS-denied environments. *2020 28th signal processing and communications applications conference (SIU)*. Piscataway: IEEE Press; 2021. p. 1–4.
25. Wilson AN, Kumar A, Jha A, et al. Embedded sensors, communication technologies, computing platforms and machine learning for UAVs: A review. *IEEE Sens J* 2022;**22**(3):1807–26.
26. Wan X, Shao YB, Zhang SY, et al. Terrain aided planetary UAV localization based on geo-referencing. *IEEE Trans Geosci Remote Sens* 2022;**60**:1–18.
27. Guo KX, Li XX, Xie LH. Ultra-wideband and odometry-based cooperative relative localization with application to multi-UAV formation control. *IEEE Trans Cybern* 2020;**50**(6):2590–603.
28. Hemann G, Singh S, Kaess M. Long-range GPS-denied aerial inertial navigation with LIDAR localization. *2016 IEEE/RSJ international conference on intelligent robots and systems (IROS)*. Piscataway: IEEE Press; 2016. p. 1659–66.
29. Shen HM, Zong Q, Tian BL, et al. Voxel-based localization and mapping for multirobot system in GPS-denied environments. *IEEE Trans Ind Electron* 2022;**69**(10):10333–42.
30. Shen HM, Zong Q, Lu HC, et al. A distributed approach for lidar-based relative state estimation of multi-UAV in GPS-denied environments. *Chin J Aeronaut* 2022;**35**(1):59–69.
31. Xu H, Wang LQ, Zhang YC, et al. Decentralized visual-inertial-UWB fusion for relative state estimation of aerial swarm. *2020 IEEE international conference on robotics and automation (ICRA)*. Piscataway: IEEE Press; 2020. p. 8776–82.
32. Walter V, Saska M, Franchi A. Fast mutual relative localization of UAVs using ultraviolet LED markers. *2018 international conference on unmanned aircraft systems (ICUAS)*. Piscataway: IEEE Press; 2018. p. 1217–26.
33. Zhu F, Ren Y, Kong F, et al. Decentralized lidar-inertial swarm odometry. arXiv preprint:2209.06628; 2022.
34. Zhang WZ, Zhang W. An efficient UAV localization technique based on particle swarm optimization. *IEEE Trans Veh Technol* 2022;**71**(9):9544–57.
35. Gageik N, Benz P, Montenegro S. Obstacle detection and collision avoidance for a UAV with complementary low-cost sensors. *IEEE Access* 2015;**3**:599–609.
36. Yu Z, Zhang Y, Jiang B, et al. A review on fault-tolerant cooperative control of multiple unmanned aerial vehicles. *Chin J Aeronaut* 2022;**35**(1):1–18.
37. Wang YD, Sun J, He HB, et al. Deterministic policy gradient with integral compensator for robust quadrotor control. *IEEE Trans Syst Man Cybern* 2020;**50**(10):3713–25.
38. Hou YN, Liu LF, Wei Q, et al. A novel DDPG method with prioritized experience replay. *2017 IEEE international conference on systems, man, and cybernetics (SMC)*. Piscataway: IEEE Press; 2017. p. 316–21.
39. Rastogi D. Deep reinforcement learning for bipedal robots [dissertation]. Delft: Delft University of Technology; 2017.

40. Hu JW, Wang LH, Hu TM, et al. Autonomous maneuver decision making of dual-UAV cooperative air combat based on deep reinforcement learning. *Electronics* 2022;**11**(3):467.
41. Zhang JD, Yang QM, Shi GQ, et al. UAV cooperative air combat maneuver decision based on multi-agent reinforcement learning. *J Syst Eng Electron* 2021;**32**(6):1421–38.
42. Lin Y, Wang MY, Zhou XL, et al. Dynamic spectrum interaction of UAV flight formation communication with priority: A deep reinforcement learning approach. *IEEE Trans Cogn Commun Netw* 2020;**6**(3):892–903.
43. Zhang YZ, Wu ZR, Ma YH, et al. Research on autonomous formation of Multi-UAV based on MADDPG algorithm. *2022 IEEE 17th international conference on control & automation (ICCA)*. Piscataway: IEEE Press; 2022. p. 249–54.
44. Abadi M, Barham P, Chen J, et al. Tensorflow: a system for large-scale machine learning. *12th {USENIX} symposium on operating systems design and implementation ({OSDI} 16)*. Berkeley: USENIX Association; 2016. p. 265–83.
45. Pham H, La H, Feil-Seifer D, et al. Autonomous UAV navigation using reinforcement learning. arXiv preprint: 1801.05086, 2018.
46. Pham H, La H, Feil-Seifer D, et al. Cooperative and distributed reinforcement learning of drones for field coverage. arXiv preprint:1803.07250, 2018.
47. Guo YH, Chen G, Zhao T. Learning-based collision-free coordination for a team of uncertain quadrotor UAVs. *Aerosp Sci Technol* 2021;**119**:107127.
48. Hu JY, Niu HL, Carrasco J, et al. Fault-tolerant cooperative navigation of networked UAV swarms for forest fire monitoring. *Aerosp Sci Technol* 2022;**123**:107494.
49. Hu ZJ, Gao XG, Wan KF, et al. Relevant experience learning: a deep reinforcement learning method for UAV autonomous motion planning in complex unknown environments. *Chin J Aeronaut* 2021;**34**(12):187–204.
50. Zhou WH, Li J, Liu ZH, et al. Improving multi-target cooperative tracking guidance for UAV swarms using multi-agent reinforcement learning. *Chin J Aeronaut* 2022;**35**(7):100–12.